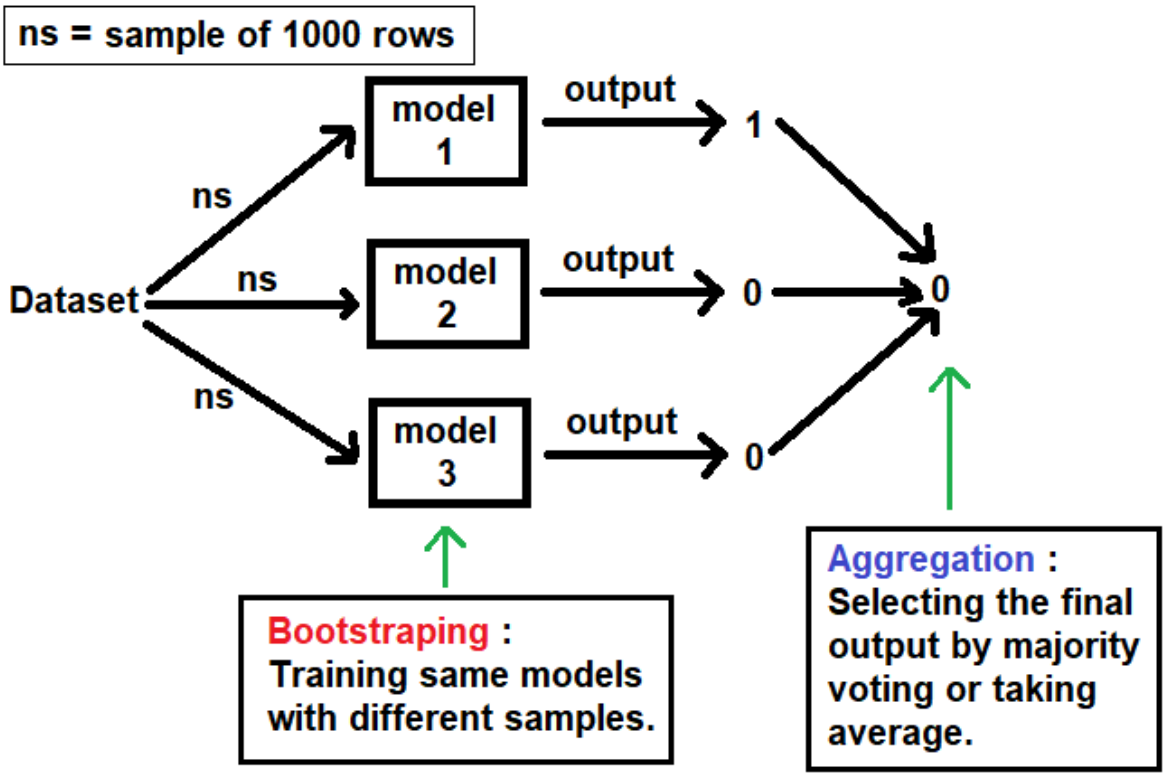
Introduction

Bagging : Where you train the same models with different samples from the dataset. Bagging = Bootstrapping + Aggregation.

Bootstrap means sampling WITH REPLACEMENT.



Why Bagging works?

We give Bagging **Low Bias, High Variance** same models and bagging returns us **Low Bias, Low Variance** result. How?

A model is Low Bias and High Variance denotes Overfitting, performs good on training data but bad on testing data. Why? Because the original data has noises in it. So when we gave 1000 different rows (with or without replacement doesn't matter) to each model, we were also splitting the noise, right? Now one model will have the majority variance, then other models will have comparatively lower variance which means other models would return output made from lower variance input. So ultimately we'll have an output that is low bias which it was before and low variance now due to reduction of variance in difference samples.

Types of Bagging

In bagging we extract the sample|rows with or without replacement e.g. df.sample(n = 1000, replace = True).

1) **Pasting** : Row sampling without replacement. df.sample(replace = False).
2) **Random Subspaces** : Column sampling with or without replacement e.g. df.sample(n = 3, axis = 1)
3) **Random Patches** : When you want to do both row and column sampling with or without sampling e.g. df.sample(n = 1000, replace = True).sample(n = 3, axis = 1).

Why is Bagging better than Pasting?

Because Bagging has more diversity and thus less variance which is totally opposite in Pasting.

How does bagging have more diversity? (Below explanation is from chatgpt)

Imagine you and a friend each build a little decision tree on four numbers drawn from the same set: 10, 20, 30, 40, 50.

With pasting (no replacement), your “bags” look like random 4-element picks without repeats.

- Example bag A: 10, 20, 30, 40.
- Example bag B: 10, 20, 40, 50.
- Notice both bags contain four *different* values, and both look pretty much like the original list minus one item.

With bagging (with replacement), your bags allow repeats and omissions.

- Example bag A: 10, 10, 10, 20.
- Example bag B: 20, 20, 30, 50.
- Now look: bag A is hugely “biased” toward 10 (three copies!), and drops 30, 40, 50 entirely. Bag B is biased toward 20, and drops 10, 40.

Because each tree in bagging sees wildly different frequency patterns—sometimes one number is shouted out three times, other numbers are completely silenced—each tree “learns” in a more unique way. In pasting, though the exact combination of four numbers changes, every tree still sees four *distinct* values close to the original distribution, so their learned rules tend to be more similar.

For large datasets in bagging, each model has a group of repeated values and in pasting each model has a unique set of values but any model can have a subset of values from another model. (Inspired from the comment of DataTalesByMuskan of Campusx tutorial)

It's like – (My own explanation)

Bagging : A building where each floor has a different amount of large rooms.

Pasting : A building where each floor has the same amount of large rooms.

So which building has more variations? **Bagging**!

How bagging returns lower variance output :

Bagging :

We can clearly say now that the models in Bagging return more diverse results especially in regression :

- One says 100, one says 80 → average = 90.
- Add more trees: say they predict 95, 85, 92 → overall average ≈ 90.4.
- All those ups and downs cancel each other out, pulling you toward the true signal.

- **Diverse mistakes**: Because each tree sees a different random mix (repeats + omissions), their mistakes aren't all in the same direction.
- If one tree's error is +10, another's is –8, another's is +5, another's is –7:
- - **Average error**: (10 – 8 + 5 – 7) / 4 = 0 — those ups and downs cancel, so your overall error is much smaller.

Pasting :

- **Correlated Mistakes :** Because trees see nearly the same distribution, their prediction errors tend to "pull" in the same direction.
- Example: four trees predict {100, 102, 101, 99} → errors {+5, +7, +6, +4}.
- Average error = (5 + 7 + 6 + 4)/4 = +5.5.
- Even though you're averaging, all errors are positive, so you don't cancel them out as effectively.

So pasting still reduces variance a bit (because each tree sees a slightly different slice), but nowhere near as dramatically as bagging's "with-replacement" shuffling.

Final Judgement :

- Since models in Bagging return a variety of errors of prediction, they tend to cancel out each other as much as possible when you average their predictions and hence the final average output by the bagging ensemble having lower error returns lower variance.

- Each model sees almost the same data slice, so their errors are similar (correlated) and don't cancel out and hence Pasting returns the final output that has much error, can cause overfitting.